STUMPRIDER · FEASIBILITY ASSESSMENT

StumpRider feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stumprider		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 (SMR-DEC-001 O6 to O9) written in; O10 back with Amish
0.1	2026-10-24	Amish Chadha	First issue in the kit 1.9.1 upgrade, on the TRL 3 design (option A)

StumpRider feasibility assessment

StumpRider is feasible on paper as a village-made hand tool: every part is cut, bent, drilled and stick-welded from common stock, the physics of the pin, the canoe heel and the freeing pull sit within known science (SMR-STM-001), and the kit's parts cost about USD 95.50 (aluminium prototype) or USD 66.90 (bamboo local variant). Two things are not shown on paper and decide whether it is worth making: whether the ring and hook head free real snags (PR-001, at risk) and whether local prices can bring a kit near USD 40 (PR-010, not met). Method: STANDARDS section 23.

> **Safety:** Feasibility here never includes anyone entering the water. The heel limits, the 7 m rule, the calibrated pin and life jackets are conditions of every trial.

Table 1. Feasibility.

Aspect	Assessment	Rating	Risks and notes	Links
Design	A hinged ring, a forked pole head on a shear pin and a three-section pole; constructability checks pass (144 of 144 in `cad/src/model.py --check`) and the D3 model passes `cad_check.py` with 77 parts and no interference	High	Hook head fixed to 12 mm bar (F1); dynamic heel limit open (O6)	SMR-STM-001; SMR-CAL-001 s6; `cad/fidelity.json`
Materials	Mild steel bar, flat and tube; 6063 aluminium tube or treated bamboo; soft aluminium wire; HDPE, polyester cord, closed-cell foam	High	O-temper pin wire may be hard to find; hard-drawn electrical wire is far stronger and must not be used (PR-011)	SMR-STM-001 S1; `bom/bom.csv`
Manufacturing and assembly	Village smith with a stick welder, drill and vice; about 8.1 h a kit; galvanising sent out	Medium	PR-009 at risk by minutes; ring roundness within 2 mm to confirm (S17); galvaniser access	SMR-CAL-001 s9; SMR-BLD-001
Power	None: hand tool	High	None	
Controls	None; the shear pin is the only safety device	Medium	The pin works only if every reel is calibrated and no substitute pin is used	PR-011; SMR-BLD-001 safety stop S2
Function	Hooked nets need 81 to 126 N against a 256 N pin floor; wrapped nets need the hook head to work them back	Medium	Friction and how nets lie on real branches unknown; staged-snag trials at TRL 4 decide	PR-001; SMR-CAL-001 s4
Safety	Static heel 3.4 deg in the design canoe; small dugouts excluded; jerk peak 6.5 deg	Medium	Steady-lift rule, 5 deg static limit (O6) and measured displacement x GM (O9), decided 2026-10-25; heel tests by canoe class at TRL 4	PR-003; SMR-RSK-001
Supply	Steel and hardware are local; aluminium tube and pin wire may come from Accra or Kumasi; bamboo is local	Medium	Local prices to confirm (DEC items to confirm 6)	SMR-MKT-001
Cost	USD 95.50 aluminium prototype, USD 66.90 bamboo local variant against USD 40	Low	PR-010 not met at prototype prices	SMR-CAL-001 s9, s10
Partner	No co-design partner approached yet	Low	Everything at TRL 4 depends on one	SMR-PRB-001 co-design; SMR-PLN-001